

① In computer, according to IEEE 754 single precision-floating representation, a 32-bit float is stored as

sign - bit 31

exponent - bits 30-23

mantissa - bits 22-0

→ Exponent is $E-127$ since it starts from bit 30-23. So, it has a bias of 127.

→ Mantissa is $\left(1 + \frac{M}{2^{23}}\right)$

→ Floating point number $x = \left(1 + \frac{M}{2^{23}}\right) \cdot 2^{(E-127)}$

Similarly integer representation of 32-bit number is

$$I_x = 2^{23}E + M$$

$$\frac{I_x}{2^{23}} = E + \frac{M}{2^{23}}$$

→ Now take $\log_2 x$ on floating point representation:

$$\log_2 x = \underbrace{\log_2 \left(1 + \frac{M}{2^{23}}\right)}_{\downarrow} + E - 127$$

$$\frac{M}{2^{23}} + \mu \quad \text{for } 0 \leq M < 2^{23}$$

$$\log_2 x \approx \frac{M}{2^{23}} + \mu + E - 127$$

$$\log_2 x = \frac{I_x}{2^{23}} - 127 + \mu \quad \left(\because I_x = E \cdot 2^{23} + M \text{ integer rep} \right)$$

$$\rightarrow y = \frac{1}{\sqrt{x}}$$

$$\log_2 y = -\frac{1}{2} \log_2 x$$

$$\frac{I_y}{2^{23}} + \mu - 127 \approx -\frac{1}{2} \left(\frac{I_x}{2^{23}} - 127 + \mu \right)$$

$$\frac{I_y}{2^{23}} = -\frac{1}{2} \frac{I_x}{2^{23}} + 127 - \mu - \frac{1}{2} (\mu - 127)$$

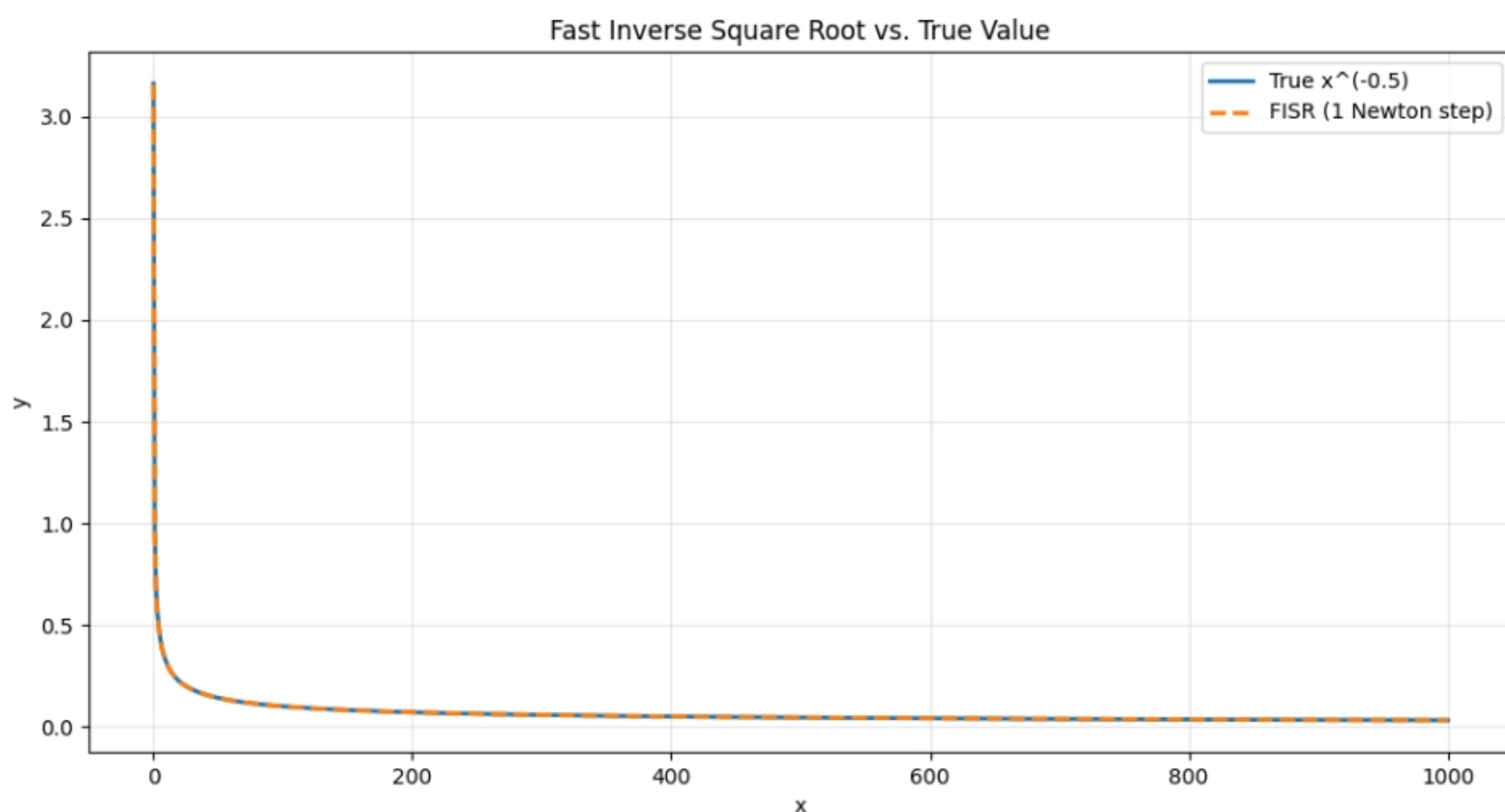
$$\frac{I_y}{2^{23}} = -\frac{1}{2} \frac{I_x}{2^{23}} + \frac{3}{2} (127 - \mu)$$

$$\overset{y_{\text{long}}}{\swarrow} \frac{I_y}{2^{23}} = \underbrace{\frac{3}{2} \cdot 2^{23} (127 - \mu)}_{\downarrow} - \frac{1}{2} I_x \xrightarrow{x_{\text{long}}}$$

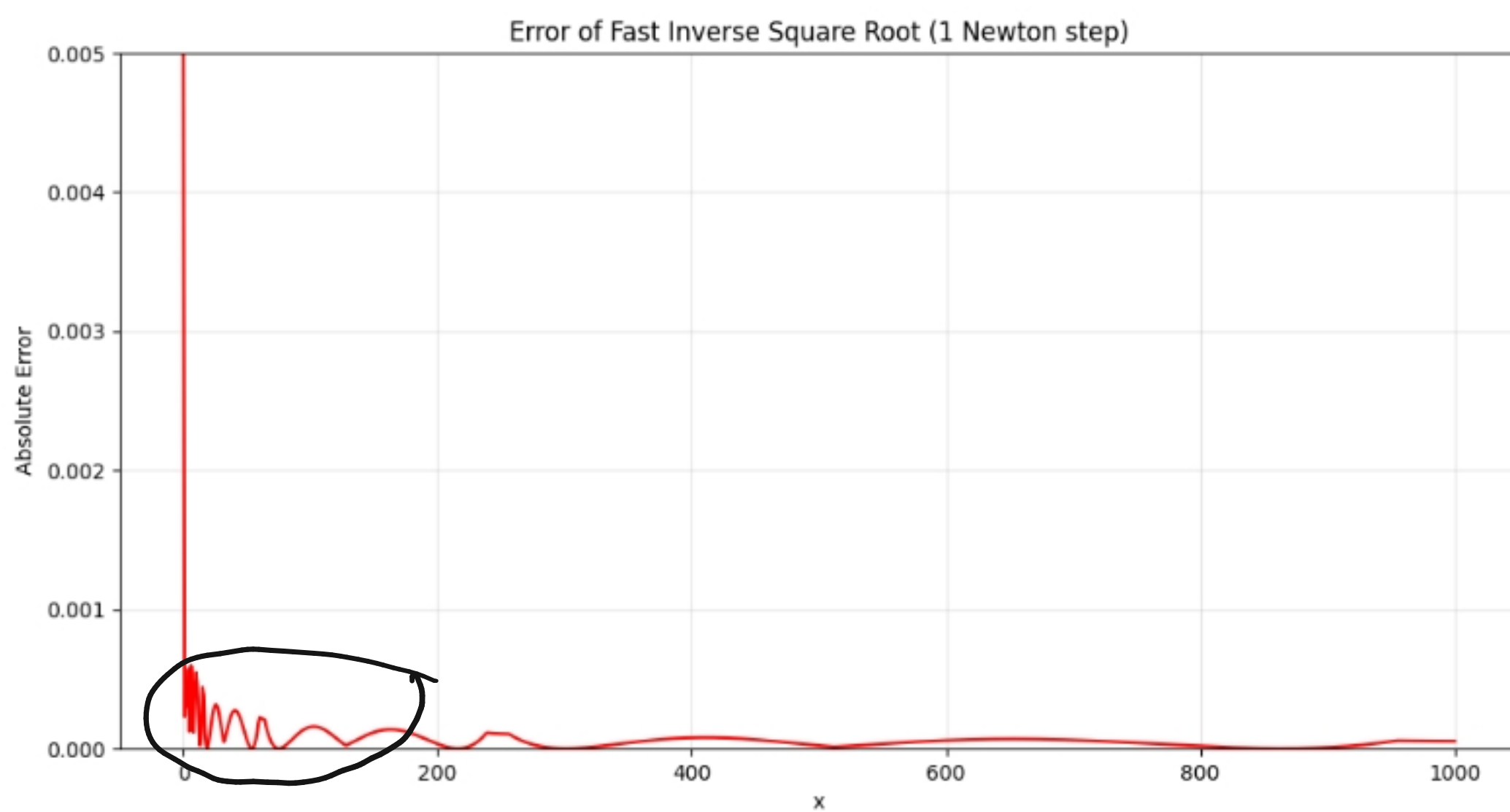
$$\underline{0x5f3759df} \quad \& \quad \mu = 0.0450466$$

↓
This magic constant is an integer when represented in IEEE 754 32-bit floating point number gives good initial approximation of $1/\sqrt{x}$.

② Plot of FISR vs true value of sqrt of x over 0-1 to 1000.

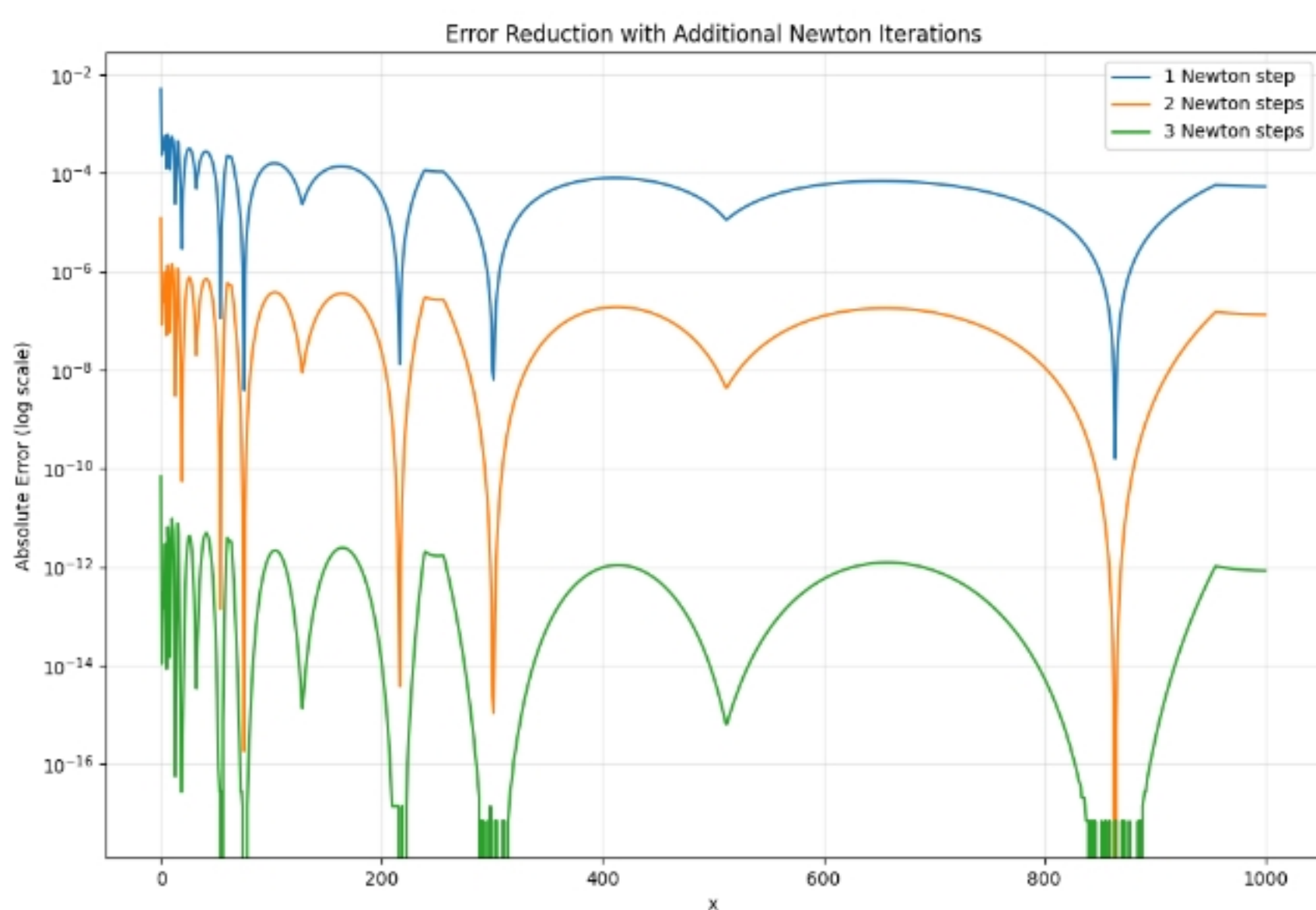


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The error is high around 0-200 range and it decreases as x increases. There is a slight increase in error value between 200-300 and remains consistently low thereafter.

4



When comparing in log-scales, the error consistently decreases with more numbers of Newton steps refinement.

→ With 1 step, the log-error is around 10^{-4} to 10^{-3} .

→ With 2 steps, it is around 10^{-7} to 10^{-6} .

→ With 3 steps, it is around 10^{-14} to 10^{-12} .